HOMEWORK 2, WEEK 1

$f_1(p) = p(-5) + p(5)$

$f_2(p) = p_1'(5) + p''(0) \qquad \mathbb{R}_2[x] \to \mathbb{R}$

$f_3(p) = \int_{-1}^{1} p(x)\,dx$

$\varphi(p,q) = \int_0^1 p(x)q(x)\,dx \qquad \mathbb{R}_2[x] \times \mathbb{R}_2[x] \to \mathbb{R}$

① $f_1, f_2 \in \mathbb{R}_2[x]$ are linear forms because sum, first and second derivative are linear operations

② · additivity $f_3(p+q) = \int_{-1}^{1}(p+q)(x)\,dx = \int_{-1}^{1}(p(x)+q(x))\,dx =$

$= \int_{-1}^{1} p(x)\,dx + \int_{-1}^{1} q(x)\,dx = f_3(p) + f_3(q)$

· homogenity $f_3(\alpha p) = \int_{-1}^{1}(\alpha p)(x)\,dx = \int_{-1}^{1}\alpha\, p(x)\,dx = \alpha\int_{-1}^{1} p(x)\,dx = \alpha f_3(p)$

$\to$ linear

③ $\dim \mathbb{R}_2[x] = 3$. To prove $\{f_1, f_2, f_3\}$ are basis of $(\mathbb{R}_2[x])^*$ with the same $\dim = 3$, we have to show they are independent.

Let $p(x) = ax^2 + bx + c$ in general form.

$f_1(p) = 50a + 2c$

$f_2(p) = 12a + b$

$f_3(p) = \frac{2}{3}a + 2c$

$\alpha_1 f_1(p) + \alpha_2 f_2(p) + \alpha_3 f_3(p) = 0$

... rewrite as

$a(50\alpha_1 + 12\alpha_2 + \frac{2}{3}\alpha_3) + b(\alpha_2) + c(2\alpha_1 + 2\alpha_3) = 0$

this has to hold for every polynomial, so we can set $b=1$ and $a, c = 0 \to \alpha_2 = 0$

similarly $\alpha_1 = \alpha_3$ (from $c=1$) and $\alpha_1 = 0$ (from $a=1$)

because the only solution where a linear combination of linear forms equals 0 is trivial, they are independent. Their number matches dimension of the space so they form a basis.

④ for $\beta^* = \{f_1, f_2, f_3\}$ to be the dual basis to $\beta = \{p_1, p_2, p_3\}$ it has to hold:

$$f_i(p_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad *$$

This problem can be written in matrix form.

$$F = \begin{bmatrix} f_1(1) & f_1(x) & f_1(x^2) \\ f_2(1) & f_2(x) & f_2(x^2) \\ f_3(1) & f_3(x) & f_3(x^2) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 50 \\ 0 & 1 & 12 \\ 2 & 0 & \frac{2}{3} \end{bmatrix}$$

$$\text{unknown } P = \begin{array}{c} \\ \\ \\ \\ \end{array} \begin{array}{c} \begin{bmatrix} c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{bmatrix} \\ \begin{array}{ccc} p_1 & p_2 & p_3 \end{array} \end{array} \begin{array}{c} 1 \\ x \\ x^2 \\ \\ \end{array}$$

$*$ states $F \cdot P = I$

$P = F^{-1}$

solution calculated in Python

$$P = \begin{bmatrix} -\frac{1}{148} & 0 & \frac{75}{148} \\ -\frac{9}{37} & 1 & \frac{9}{37} \\ \frac{3}{148} & 0 & -\frac{3}{148} \end{bmatrix} \sim \begin{array}{l} p_1(x) = \frac{3}{148}x^2 - \frac{9}{37}x - \frac{1}{148} \\ p_2(x) = x \\ p_3(x) = -\frac{3}{148}x_1^2 + \frac{9}{37}x + \frac{75}{148} \end{array}$$

⑤ · left linearity $\ell(\alpha_1 p_1 + \alpha_2 p_2, Q) = \int_0^1 (\alpha_1 p_1 + \alpha_2 p_2)(x) Q(x)\,dx =$

$= \int_0^1 (\alpha_1 p_1(x) Q(x) + \alpha_2 p_2(x) Q(x))\,dx = \int_0^1 \alpha_1 p_1(x) Q(x)\,dx + \int_0^1 \alpha_2 p_2(x) Q(x)\,dx =$

$= \alpha_1 \int_0^1 p_1(x) Q(x)\,dx + \alpha_2 \int_0^1 p_2(x) Q(x)\,dx = \alpha_1 \ell(p_1, Q) + \alpha_2 \ell(p_2, Q)$

· right linearity proof is identical as product is commutative

$\rightarrow$ linear

ℓ expressed as weighted sum of tensor products:

$\ell = \sum_{i,j=1}^{3} a_{ij}(f_i \otimes f_j)$, where $a_{ij} = \ell(p_i, p_j)$ corresponding to basis β

Let $A = [a_{ij}]$ and $S = \begin{bmatrix} \ell(1,1) & \ell(1,x) & \ell(1,x^2) \\ \ell(x,1) & \ell(x,x) & \ell(x,x^2) \\ \ell(x^2,1) & \ell(x^2,x) & \ell(x^2,x^2) \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$

corresponding to standard basis $\{1, x, x^2\}$

Then $A = P^T S P$, with top left element $\frac{1037}{54760}$ ...

STATEMENT: This is my work. Denis Nedić told me about the matrix form in part 4. I used Gemini Pro to verify some results and to write the Python script.